



Scan for Portfolio

OBED MAVUNGU

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Education

Virginia Polytechnic Institute and State University

Expected May 2028

Bachelor of Engineering in Computer Science

Blacksburg, Virginia

Coursework: Data Structures & Algorithms, Computer Organization, Discrete Math, Calculus III

Award: Best Use of .Tech (MLH) at HackPSU for Syllabyte, a multi-agent AI study platform built in 24 hours

Extracurricular: NSBE, Amateur Boxer, Movie Writer

Experience

Zarnite (AI Startup)

June 2026 – Present

Full Stack Developer Intern

Remote

- On a 4-engineer team shipping a real-time social platform, wrote **52** of the **97** backend authorization tests, pinning down the video-room token path where a client-supplied room name could otherwise hand a stranger audio access to a private call, and put them on a GitHub Actions pipeline that runs on every push.
- Built the Next.js web client from the ground up, **28** of its **33** source files across **10** routes, delivering email sign-in, sign-up with automatic profile creation, likes and follows against a **65**-endpoint FastAPI backend and a **17**-table PostgreSQL schema governed by **50** row-level security policies.
- Containerized the FastAPI service and moved production from Render to Google Cloud Run through GitHub Actions, tightening CORS and room-membership checks on the real-time token endpoint in the same pass.

Auburn University of Montgomery Learning Center

August 2025 – Present

Mathematics & Computer Science Tutor

Montgomery, Alabama

- Teach data structures, algorithms and programming fundamentals to **20–25** students weekly, explaining technical concepts to non-expert audiences and sustaining **90%+** session return rates.

Technical Projects

Wapi | Restaurant Discovery Platform, Kinshasa | *React, Supabase (PostgreSQL), Capacitor, Node.js*

May 2026

- Kinshasa has no reliable restaurant directory, so I founded and solo-built one, writing an ingestion pipeline that merged the Google Places and OpenStreetMap APIs into **1,888** deduplicated venues using trigram similarity and Haversine distance.
- A large share of records carried unusable coordinates, so instead of paying for a re-scan I built a nearest-centroid reverse-geocoder over **24** commune centroids; it recovered **77%** of them and populated **22 of 24** communes at **\$0**.
- Shipped one React codebase to web (Vercel CI/CD) and mobile (Capacitor, iOS and Android), with live community feeds over Supabase Realtime.

PunchIQ | End-to-End ML Prediction System | *Python, scikit-learn, pandas, FastAPI, SHAP, React*

June 2026

- Built and deployed a two-sport fight prediction system (boxing and UFC) behind a FastAPI service, returning SHAP attributions with every prediction so a non-technical user can see which features drove the call.
- My first model scored **81%**, which was too good: career-final win-loss records already encoded the fight being predicted. I proved the leak by showing a one-line heuristic nearly matched the Random Forest, then rebuilt the pipeline point-in-time with a January 2023 temporal split and **15** differential features over **13,254** training and **3,954** test fights, reporting an honest **62.2%** accuracy at a **0.2324** Brier score.
- Trimmed the model to **50** trees at depth **12** and the service to a **6.5 MB** footprint so inference runs on a free **512 MB** instance.

AI Website Builder | Full-Stack LLM SaaS | *React, Node.js, Express, PostgreSQL, Prisma, Stripe*

December 2025

- Users describing a website in plain English had to wait on a blocking request with nothing on screen, so I streamed the generated code over Server-Sent Events; the page now renders token by token while it is being written.
- Payment webhooks arrive duplicated and out of order, which would grant or lose a customer's credits, so I made the Stripe flow signature-verified and idempotent: credits are granted exactly once per confirmed payment.
- Designed and built the backend, **16** REST endpoints over an **8**-table PostgreSQL schema, powering authenticated accounts, project versioning, one-click rollback and a chat-based AI editing loop.

Technical Skills

Languages: Python, JavaScript, Java, C++, C

Frameworks: React, Next.js, Node.js, Express, FastAPI, Tailwind CSS

Data & Cloud: PostgreSQL, MongoDB, Supabase, Prisma, Docker, Google Cloud Run, Vercel

Machine Learning: scikit-learn, pandas, NumPy, SHAP, leakage detection, model benchmarking

Practices: REST API design, pytest, CI/CD (GitHub Actions), code review, Agile sprints

Fundamentals: Data Structures & Algorithms, OOP, WebSockets, Server-Sent Events, JWT auth